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Supplementary information

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A RISC-V 32-Bit Microprocessor Based on Two-dimensional Semiconductors

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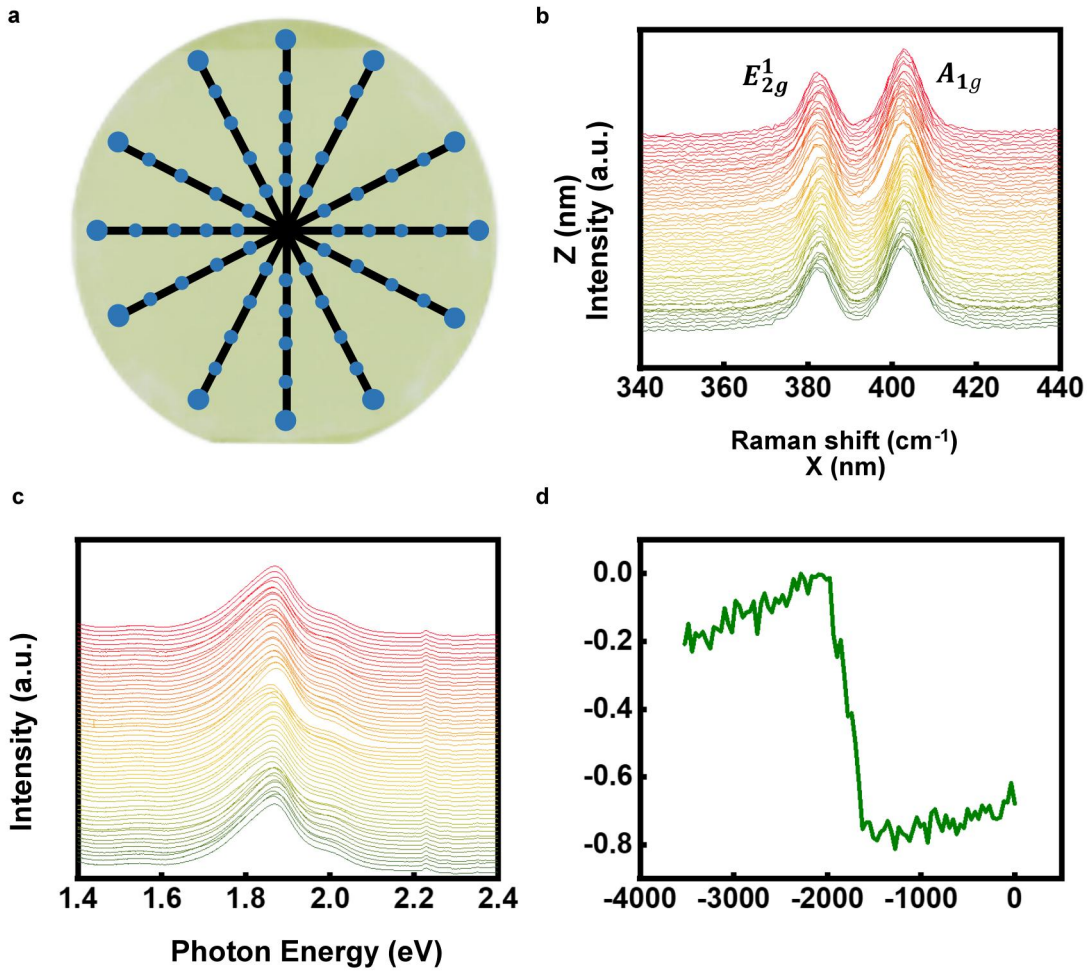
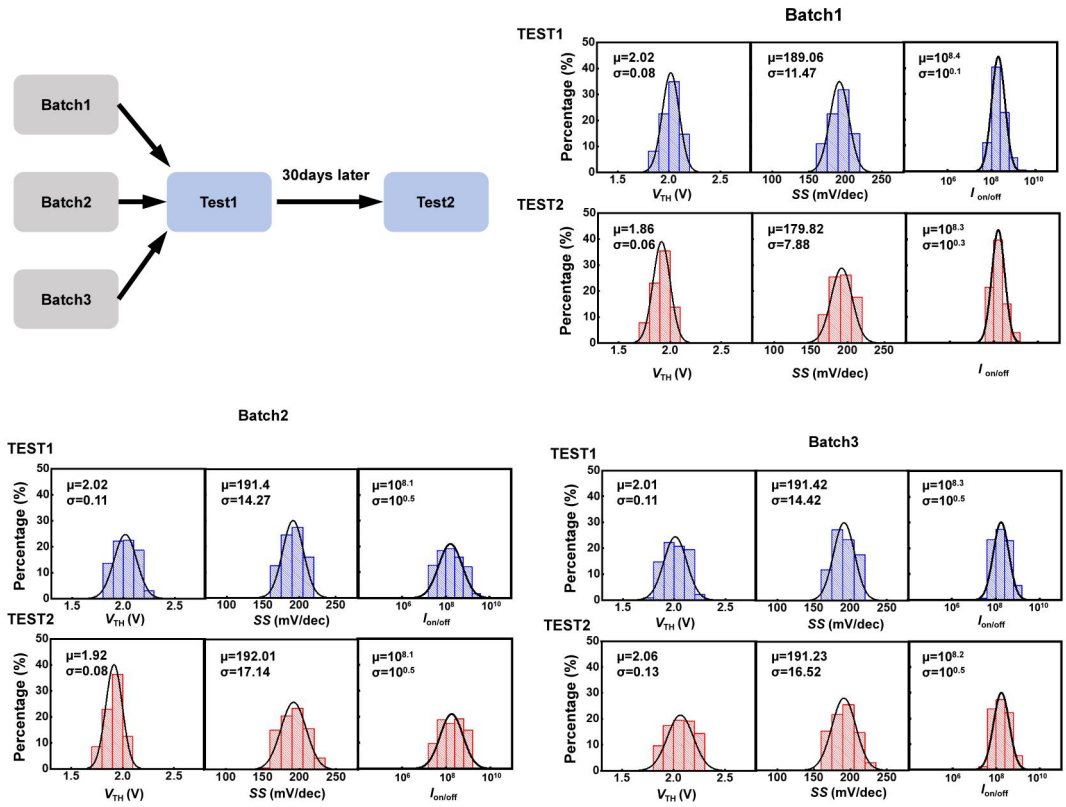
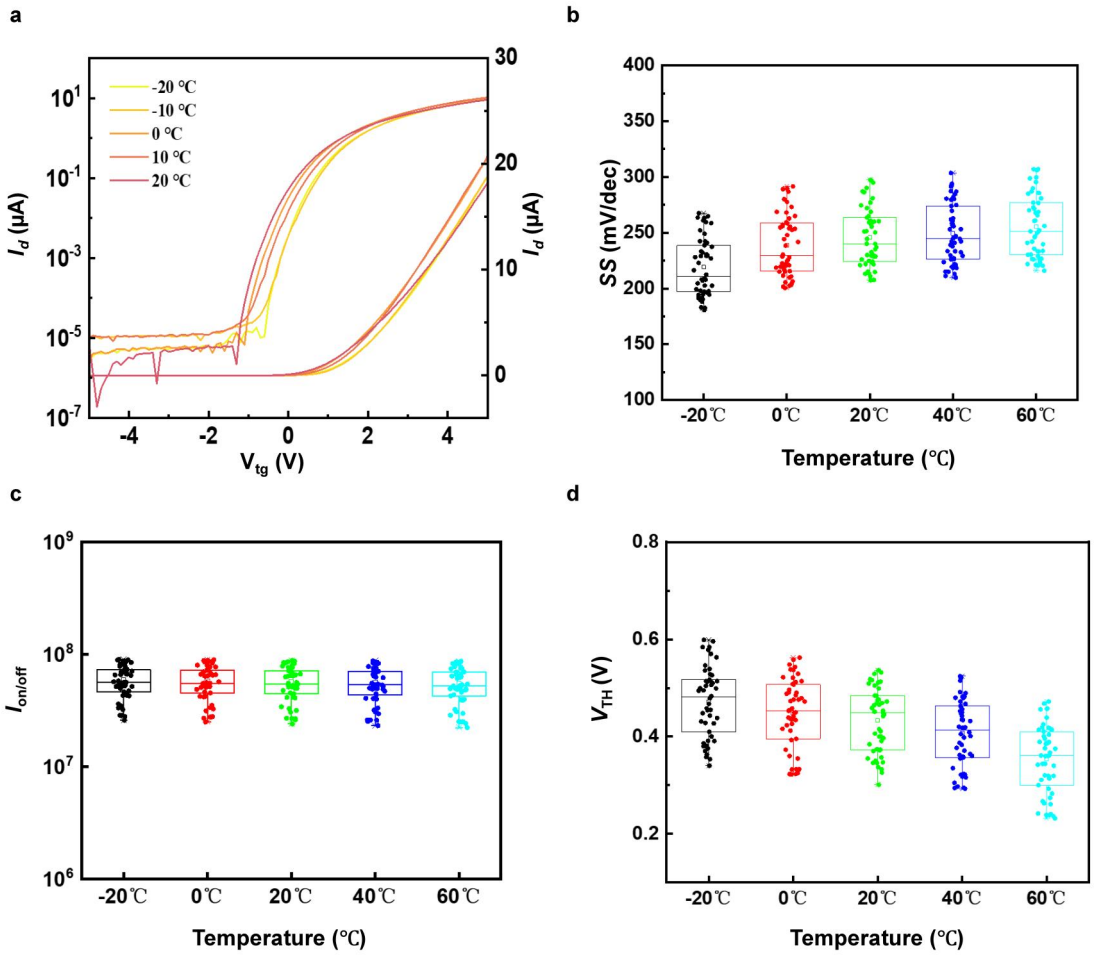
1. Supplementary Figures



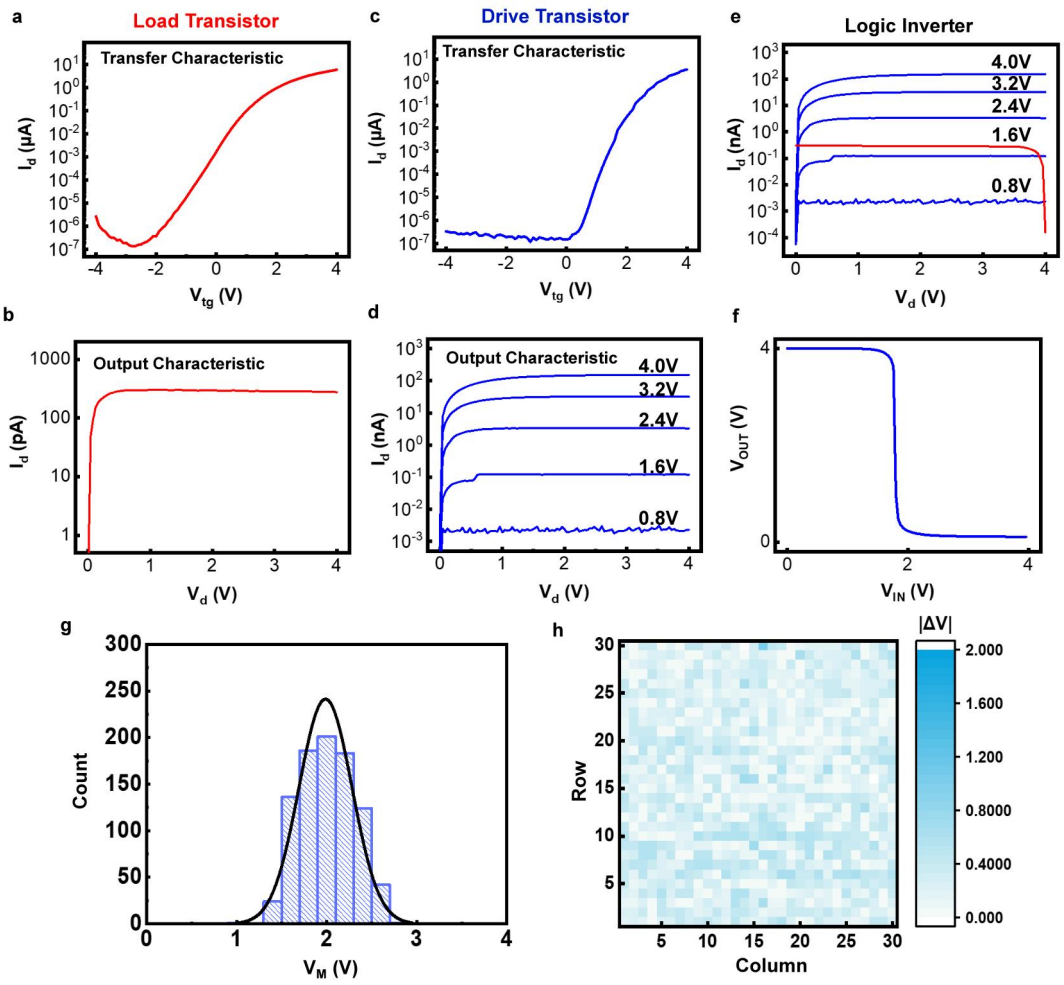
Supplementary Fig. 1| Material characterizations of wafer-scale MoS_2. a, Raman and photoluminescence tests were conducted on 60 points uniformly across a 4-inch MoS_2 thin film. **b,c,** The Raman and photoluminescence spectra, respectively, were acquired from 60 points on an as-synthesized 4-inch MoS_2 wafer. **d,** The thickness of the MoS_2 wafer was determined to be a monolayer layer using atomic force microscopy.



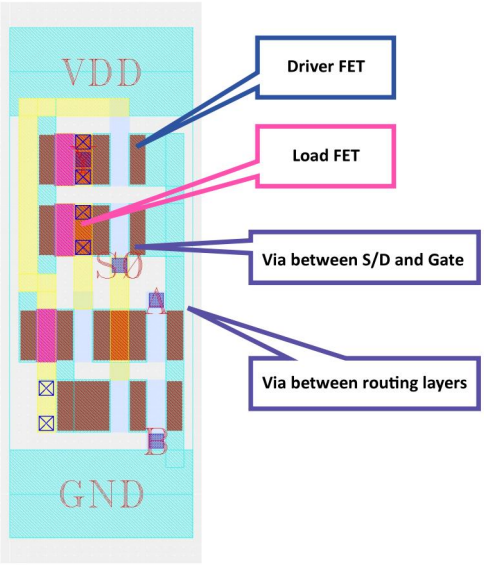
Supplementary Fig. 2| Statistical results of multi-batch wafer-level transistors. Each batch comprises 7000 transistors, among which 800 were randomly selected for testing, encompassing the V_{TH}, SS, and I_{on}/I_{off} distributions after fabrication. A second round of testing (Test 2) was conducted after a one-month interval, and the results of which substantiate the stability of the integration process and the satisfactory retention of performance over time



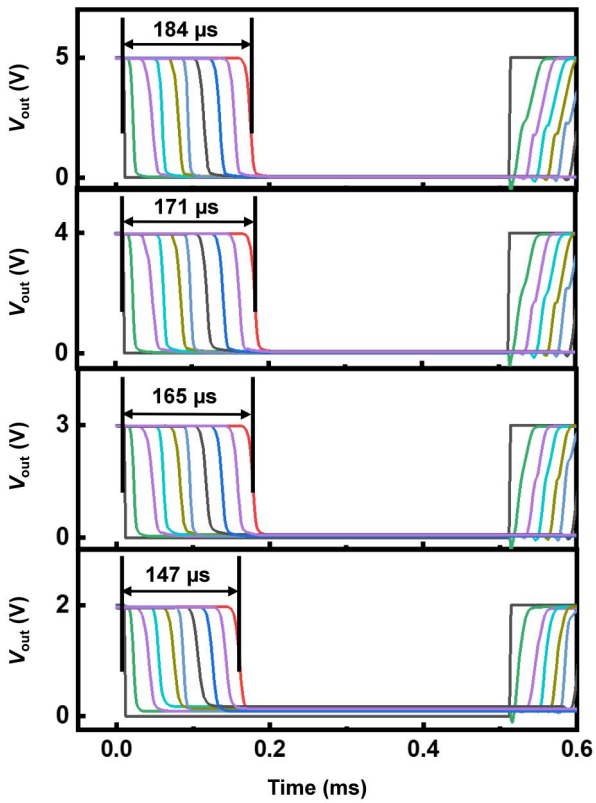
Supplementary Fig. 3| (a) Temperature-dependence of a top-gate MoS_2 transistor. (b,c and d) The temperature variation characteristics of 50 transistors for V_{TH}, SS, and I_{on}/I_{off}, respectively.



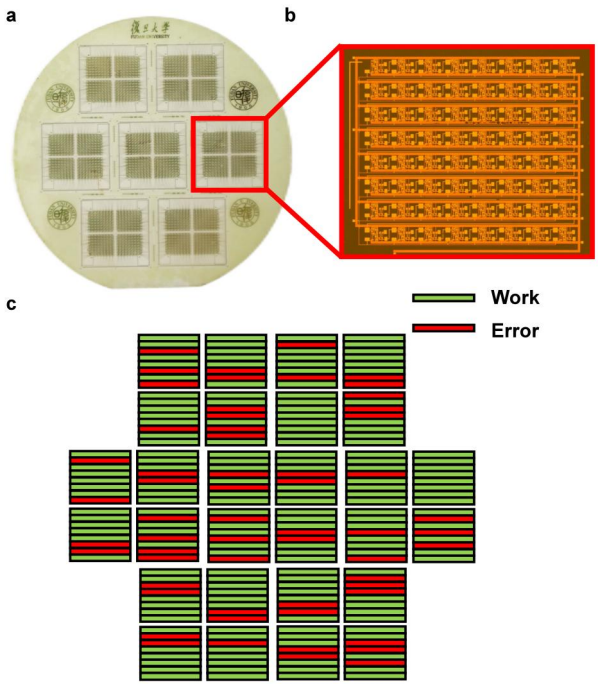
Supplementary Fig. 4| Electrical characterization of the MoS_2 inverter arrays. The transfer and output characteristics of a typical load transistor (**a-b**) and driver transistor (**c-d**) that make up a logic inverter. The load transistor's output characteristic curve was measured with a gate voltage of 0V, while the driver transistor's curve was obtained with gate control voltages ranging from 0.8V to 4.0V. **e**, The combination of the output characteristic curves of both load and driver transistors. **f,** The measured inverter voltage transfer characteristic. **g**, The statistical distributions of the V_M for 898 inverters, as well as the spatial distribution of ΔV (**h**), which is calculated as the difference between V_{dd} /2 and V_M for each of these inverters.



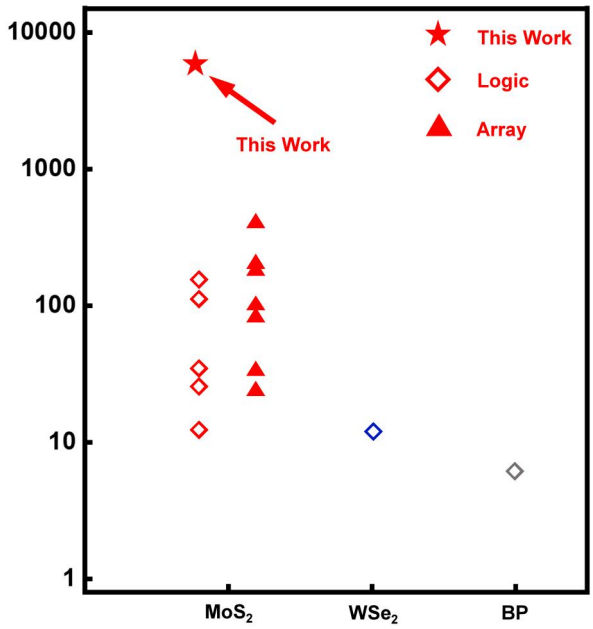
Supplementary Fig. 5| A detailed standard cell layout of an MX2 module. The layout is derived using our '2D_NMOS_lib' cell library. Different structures are used to generate a logic module, including driver FET, load FET, Via between S/D and Gate, and Via between routing layers.



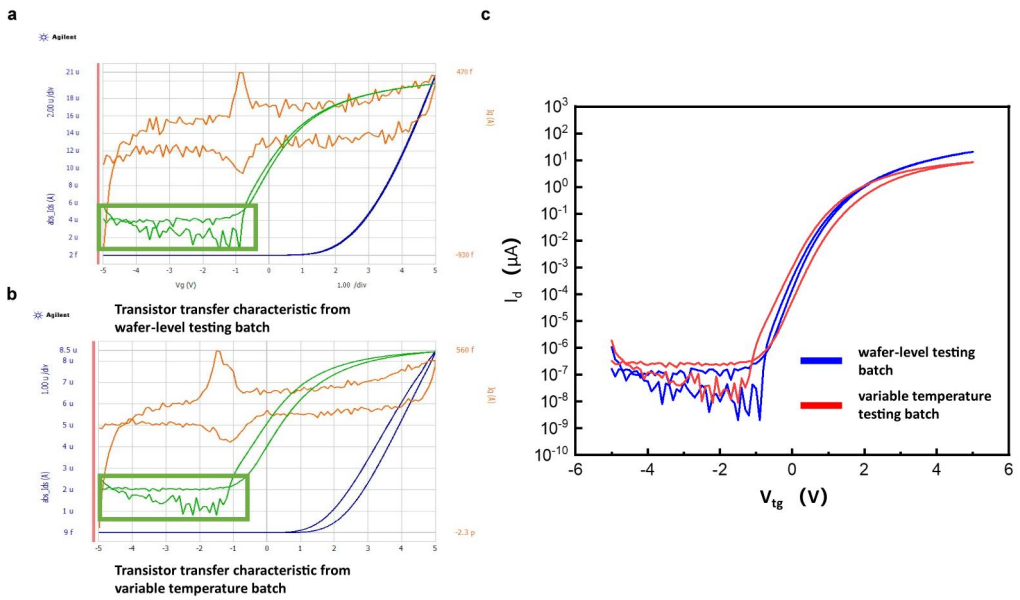
Supplementary Fig. 6| The delay of the longest path under different operating voltages. We demonstrate the delay of the longest logic gate path of our MoS_2 chip under different operating voltages.(2V-5V)



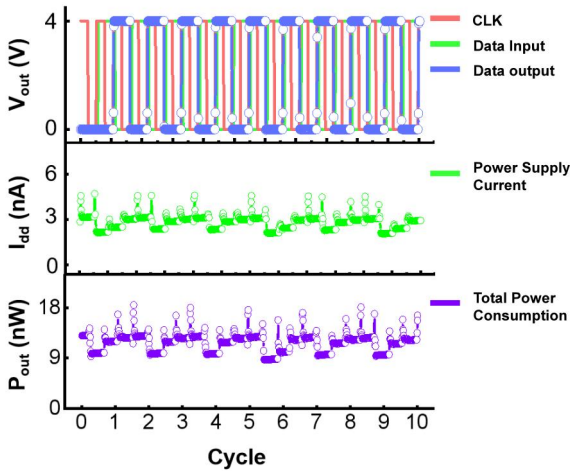
Supplementary Fig. 7| Yield analysis of wafer-level 8-bit registers. (a) A photograph of a 4-inch wafer composed of 28 matrices of 64-bit MoS2 registers, totaling 224 8-bit shift registers. **(b)** The optical microscope image of an as-manufactured 64-bit MoS2 register matrix, including 8 parallel groups of 8-bit registers. **(c)** The yield of 8-bit registers, each consisting of 144 transistors, can reach 71% on the wafer.



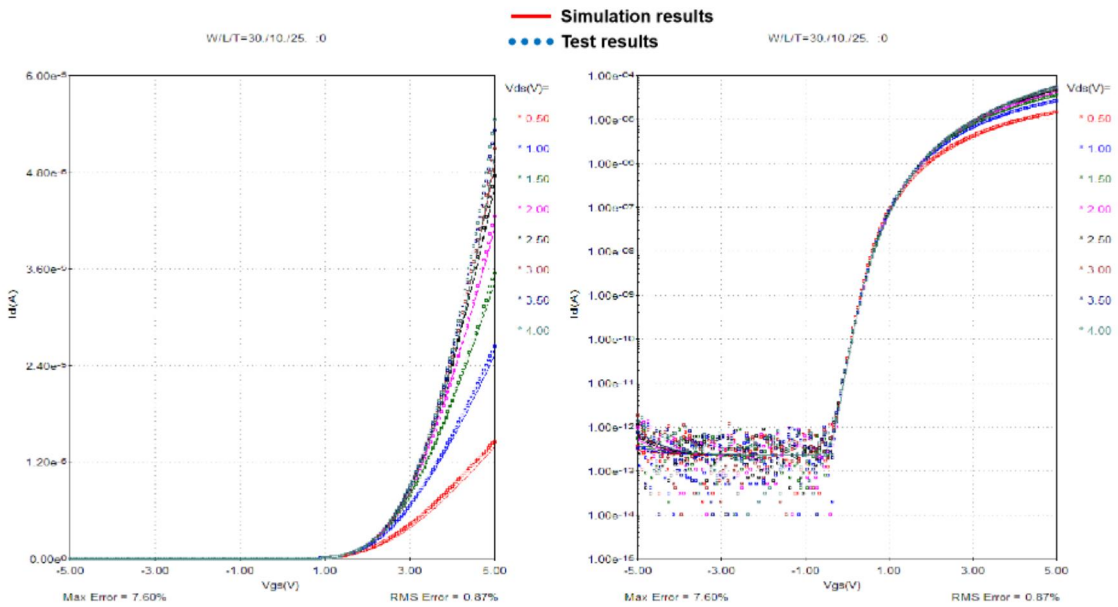
Supplementary Fig. 8| Benchmark of the integration level of ICs implemented with 2D semiconductors.



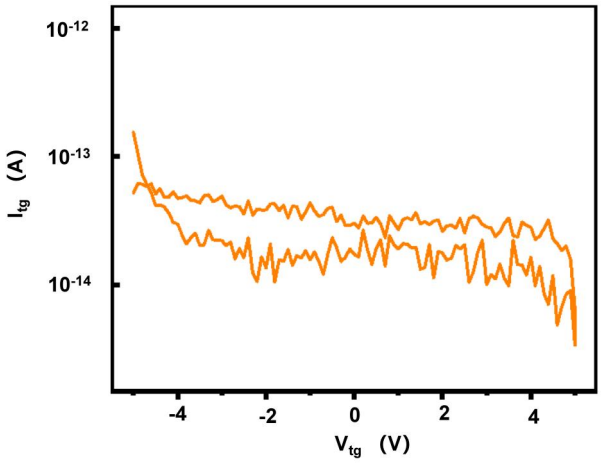
Supplementary Fig. 9| a-b. Transistor transfer characteristic from two different batches of MoS_2 films, **c.** Direct comparison of test results under two different test batches with an order of magnitude difference in baseline noise.



Supplementary Fig. 10| The input and output curves from the register, the output current and the calculated output power, all voltage and current data are obtained directly from the semiconductor analyzer.



Supplementary Fig. 11| Comparison between modelling and experimental data. **a.** shows transfer characteristics in linear coordinates and **b.** in logarithmic coordinates.



Supplementary Fig. 12| The experimental test results of the gate leakage current.

2. Supplementary Table

Supplementary Table. 1| Comparison of Circuit Performance Based on MoS2 Semiconductor Material

References	Material	Channel Length (μm)	Avg：Mobility ($cm^2/V·s$)	Avg：Vth (V)	Avg：SS (mV/dec)	Avg：Ion/Ioff
1	MoS_2	2	3	0.65	n/a	1×10^8
2	MoS_2	15	88	2.47	n/a	1×10^8
3	MoS_2	4.2	53	-1	226.4	1.52×10^8
4	MoS_2	5	6	n/a	n/a	8.13×10^4
5	MoS_2	10	70	0.96	83.1	5×10^7
6	MoS_2	7	0.3	-1.75	n/a	3.5×10^3
7	MoS_2	10	46	1.6	260	1×10^8
8	WSe_2	10	25	-32	n/a	1×10^6
9	BP	1	24	2.54	485	6.4×10^7
This Work	MoS_2	3	28	1.8	100~160 (Due to batch variations)	1×10^9 (Due to batch variations)

The last row is our work, demonstrating the performance comparison of the statistical results for MoS_2 transistors.

Supplementary Table. 2| Comparison of Processor Performance among Several New Types of Semiconductor Materials

Feature	WUJI	1-bit microprocessor [10]	PlasticARM [11]	Vector–matrix multiplier [12]
Area (mm^2)	6 × 6	0.6	59.2	n/a
Technology	3um metal-oxide TFT	2um metal-oxide TFT	0.8μm metal-oxide TFT	3um floating-gate FET
Logic type	D-mode load	D-mode load	Unipolar n-type resistive load	Cross-bar
Supply voltage (V)	4	5	3	10 (max operating voltage)
Chip pin count	207	11	28	145
Processor	32-bit RISC-V microprocessor	1bit custom microprocessor	32-bit Arm Cortex-M-based SoC	32×32 vector–matrix multiplier
Number of devices	5931	115	56,340	1024
NAND2-equivalent gate count	1977	n/a	18,334	n/a
Max circuit clock frequency (kHz)	1	2-20 (estimated)	29	1 (as filter)
Power consumption (mW)	0.431 (1kHz)	0.06 (estimated)	21	n/a
Power consumption per Gate (nW)	218	1400	n/a	n/a
Channel material	MoS_2	MoS_2	IGZO	MoS_2

The first column is our performance indicators, n/a means the data was not listed in the literature.

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